

Recap.

- ① For every RV X , $\phi_X(t) = E(e^{itX}) = \begin{cases} \sum_n e^{itx} P_X(x) \\ \int_{-\infty}^{\infty} e^{itx} P_X(x) dx \end{cases}$, where i is imaginary unit.
- ② Every characteristic function uniquely matches to a distribution.
For RVs X and Y , given $X \perp Y$, then $\Psi_{X+Y}(t) = \Psi_X(t) \Psi_Y(t)$.
- ③ Markov's inequality: for RV X defined on $[0, +\infty)$ and $a > 0$, $P(X \geq a) \leq \frac{E(X)}{a}$.
Chebyshev's inequality: for any RV X and $b > 0$, $P(|X - E(X)| \geq b) \leq \frac{\text{Var}(X)}{b^2}$.
- ④ "Given a sample of independent and identically distributed value $X_1, X_2, \dots, X_n$, average value of sample $\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$ converges to expected value $E(X_i) = \mu$."
w.l.n (converges in probability): $\bar{X}_n \xrightarrow{P} \mu$, i.e. for $\forall \epsilon > 0$, $P(|\bar{X}_n - \mu| > \epsilon) \rightarrow 0$ when $n \rightarrow \infty$.
s.l.n (almost surely converges): $\bar{X}_n \xrightarrow{a.s.} \mu$, i.e. $P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$.

Exercises.

Ex 1. Hint: Taylor series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

$$\begin{aligned} \text{For } Z \sim \text{Poi}(\lambda), \phi_Z(t) &= \sum_{n=0}^{\infty} e^{itn} \cdot P(Z=n) \\ &= \sum_{n=0}^{\infty} e^{itn} \frac{e^{-\lambda} \lambda^n}{n!} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(e^{it}\lambda)^n}{n!} \\ &= e^{-\lambda} e^{e^{it}\lambda} \end{aligned}$$

$$\phi_{X+Y}(t) = \phi_X(t) \phi_Y(t) = e^{(e^{it}-1)\lambda_1} \cdot e^{(e^{it}-1)\lambda_2} = e^{(e^{it}-1)(\lambda_1+\lambda_2)}, \quad X+Y \sim \text{Poi}(\lambda_1+\lambda_2).$$

Ex 2. Hint: For discrete RV X mapped to N and $n \in \mathbb{N}$, $P(X > n) = P(X \geq n+1)$.

- (i) $E(N_T) = E(N_A) + E(N_B) = 150$. Since $N_A \perp N_B$, $\text{Var}(N_T) = \text{Var}(N_A) + \text{Var}(N_B) = 150$.
- (ii) With Chebyshev's inequality: $P(|N_T - 150| \geq b) \leq \frac{150}{b^2}$.
 $P(N_T > 200) = P(N_T \geq 201) = P(N_T - 150 \geq 201 - 150) \leq P(|N_T - 150| \geq 51) \leq \frac{150}{51^2} \approx 5.77\%$.
- (iii) With Markov's inequality: $P(N_T > 200) = P(N_T \geq 201) \leq \frac{150}{201} = 74.63\%$.
"Variance contains more information. Chebyshev's inequality gives a tighter bound."
- (iv) $N_A \sim \text{Poi}(50)$, $N_B \sim \text{Poi}(100)$, since Expectation = Variance.
- (v) $N_T = N_A + N_B \sim \text{Poi}(150)$, $P(N_T > 200) = 1 - P(N_T \leq 200) \approx 0.0042\%$.
"Inequalities are estimations without known distribution, not precise."

Ex 3. Hint: $X_{\bar{D}^2} = \begin{cases} 1 & \text{if } (x,y) \in \bar{D}^2 \\ 0 & \text{otherwise} \end{cases}$

$$(i) P_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x,y) dy = \begin{cases} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{2}{\pi} \sqrt{1-x^2} & \text{if } x \in [-1, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$P_Y(y) = \begin{cases} \frac{2}{\pi} \sqrt{1-y^2} & \text{if } y \in [-1, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$(ii) E(X) = \int_{-\infty}^{\infty} x \cdot P_X(x) dx = \int_{-1}^1 \frac{2x}{\pi} \sqrt{1-x^2} dx = 0. \quad (\frac{2x}{\pi} \text{ is odd function and } \sqrt{1-x^2} \text{ is even})$$

$$E(Y) = 0.$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= E(X^2)$$

$$= \int_{-\infty}^{\infty} x^2 P_X(x) dx$$

$$= \frac{2}{\pi} \int_{-1}^1 x^2 \sqrt{1-x^2} dx \quad (\text{Let } x = \sin(u), \text{ then } x' = \sin'(u), 1 = \cos(u), \frac{dx}{du} = \cos(u))$$

$$= \frac{2}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2(u) \cos(u) \cos(u) du \quad (2 \sin u \cos u = \sin 2u)$$

$$= \frac{2}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4} (\sin 2u)^2 du \quad ((\sin 2u)^2 \text{ is even function})$$

$$= \frac{2}{\pi} \times \frac{1}{4} \times 2 \int_0^{\frac{\pi}{2}} (\sin 2u)^2 du \quad ((\sin u)^2 = \frac{1 - \cos 2u}{2})$$

$$= \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \frac{1 - \cos 4u}{2} du$$

$$= \frac{1}{\pi} \left[\frac{u}{2} - \frac{\sin 4u}{8} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4}.$$

$$\text{Var}(Y) = \frac{1}{4}.$$

$$(iii) \textcircled{1} \text{ Verify } p_{X,Y}(x,y) \stackrel{?}{=} p_X(x) \cdot p_Y(y).$$

Pick $(x,y) = (0.8, 0.8)$, $p_X(x) = p_Y(y) > 0 = p_{X,Y}(x,y)$, not independent.

$$\textcircled{2} \text{ Verify } \text{cov}(X,Y) = E(X \cdot Y) - E(X)E(Y) = E(X \cdot Y) \stackrel{?}{=} 0.$$

$$E(X \cdot Y) = \iint_{\bar{D}^2} x \cdot y \frac{1}{\pi} dx dy \quad (\text{define } x = r \cos \theta, y = r \sin \theta \text{ with } r \in [0, 1], \theta \in [0, 2\pi))$$

$$= \frac{1}{\pi} \iint_{\bar{D}^2} r^2 \cos \theta \sin \theta dx dy \quad (dx dy = |\det(J)| dr d\theta = \left| \begin{pmatrix} \frac{dx}{dr} & \frac{dx}{d\theta} \\ \frac{dy}{dr} & \frac{dy}{d\theta} \end{pmatrix} \right| dr d\theta = r dr d\theta)$$

$$= \frac{1}{\pi} \int_0^{2\pi} \int_0^1 r^2 \cos \theta \sin \theta r dr d\theta$$

$$= \frac{1}{\pi} \int_0^{2\pi} \cos \theta \sin \theta d\theta \int_0^1 r^3 dr$$

$$= \frac{1}{\pi} \int_0^{2\pi} \frac{1}{2} \sin(2\theta) d\theta \int_0^1 r^3 dr \quad (\int_0^{2\pi} \sin(2\theta) d\theta = 0)$$

$$= 0, \quad \text{uncorrelated.}$$

Ex 4. Solution shown in the code form.